Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/13/2005
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Dynamics of Flame Spread in Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	4
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Wakai, Kazunori (Gifu University, Japan) Takahashi, Shuhei (Gifu University, Japan)		
Grant/Contract No.:	NCC3-842		
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	Despite intensive research on microgravity flame spread in the last decade, inspired primarily from fire safety concerns in manned space travel, a number of aspects of the flame spread problem has been identified that still remain unexplored or unresolved. Primary among those is to establish experimentally whether a spreading flame self-extinguishes in a quiescent microgravity environment, and if so, under what circumstances. Although there is some experimental evidence that for thick fuels steady spread does not occur, the phenomenon of self-extinguishment, with its importance fire-safety related consequences, deserves to be scrutinized further. Moreover, the mechanism of extinguishment remains to be addressed. For instance, we would like to answer if extinction occurs at a particular critical fuel thickness, how that thickness depends on the ambient and fuel parameters, how long does it take for a flame to extinguish, etc. To settle this important issue and address a number of related outstanding problems of flame spread, we propose four		

Task Description:	space-based experiments on flame spread over thin sheets of PMMA placed in the center of a wind tunnel in which a fully developed flow of oxygen/nitrogen mixture with prescribed average velocities can be created. The experiments are made compact by moving the fuel, spooled on two rollers, opposite to the direction of the spread to make the flame stationary. As a result, experimental parameters, especially flow velocity and fuel thickness, can be varied in a single experiment without intervention from an operator. By simply changing the location of the igniter the experiment can be re-configured for concurrent-flow spread. The project is directly related to basic understanding of flame spread important for materials selection leading to minimization of risk of ignition and fire hazard in space stations and space voyages. Besides the outstanding issues, microgravity offers a unique opportunity to link the two paradigms of flame spread - wind-aided and wind opposed configurations - by gradually reversing the direction of flow velocity in the same experiment. Moreover, the rich data set produced by the proposed experiments will help verify existing spread rate formulas and fuel development of new formulas that bring together the two regimes of concurrent-flow and opposed-flow flame spread. Currently such data is available only for flow velocities greater than the velocity induced by buoyancy driven flow. As a result flame spread formulas are unavailable or untested at the low velocity range most relevant to microgravity environment. The proposed imaging diagnostics, builds upon earlier development of emission pyrometry by the first PI, and is expected to produce improved non-obtrusive measurement of line-of-sight-averaged temperature and concentration field. The orginal goal of the proposed effort is to conduct a simple yet definitive set of experiments that, coupled with computational modeling and theoretical effort, can resolve the ten outstanding issues of flame spread. Although many of these issues involve refinement of existing models to unify different sub-regimes of flame spread, issue 3, that defines appropriate gravity level in an experiment, issues 7 and 8, that experimentally resolve the mechanism of flame extinction at quiescent environment, and issue 9, that links concurrent-flow flame spread with opposed-flow spread, are the core issues of this proposal. The simplicity of the experiment allowing a huge volume of data without any additional cost offers a unique opportunity to address the rest of the issues. The importance of fire safety in manned space travel cannot be overstated. The proposed research hopes to establish criteria under which flames, previously thought to be self propagating, extinguishes by itself. If a critical fuel thickness can be reliably predicted above which steady flames cannot be sustained over a combustible material, it will have a direct impact on how combustible materials are made fire safe in the Internal Space Station. The study of transition between wind-aided spread rate to wind-opposed spread rate will allow us to leverage the microgravity experiments to link earth bound flames, which are mostly wind-aided, to wind-opposed flames, which are theoretically better understood.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The space based experiment that was originally proposed could not materialize after we were instructed to conduct only ground-based research. The ground based data we acquired in the MGLAB of Japan and parabolic flights in Nagoya were analyzed. The data agreed reasonably well with our proposed theory of microgravity flame spread rate. Our achievements can be summarized as follows. 1. We have theoretically shown that all fuels behave as thermally thin fuel in a perfectly quiescent environment. More interestingly, we have developed an expression that predicts a critical fuel thickness, above which steady flame cannot be sustained in such an environment irrespective of the ambient conditions. Our drop tower data and previously published flammability data are consistent with these conclusions. 2.We have developed flame spread rate expressions both for thin and thick fuels in a low velocity opposed-flow environment that microgravity typically offers. These expressions, similar to de Ris spread rate formulas, are first of its kind that clearly decouples the radiative effects. We have posted a Java applet in the web site of this project, flame.sdsu.edu, which calculates spread rate in opposed-flow configuration for a variety of fuel and ambient conditions. 3. Using the microgravity flame spread rate expressions, flammability maps are derived for both PMMA and cellulosic fuels, which compare quite well with the available experimental data. It is shown that the radiative effects alone dominate the microgravity flame extinction. 4. The educational outreach site we developed (http://thermo.sdsu.edu) continues to receive great comments from users, students, educators and professionals, around the world. To date more than 2300 educators have registered to use this site. The flame tracker hardware we built at SDSU has demonstrated the feasibility of our proposed experiment in space in which a flame is made stationary by moving the fuel. We have completed the construction of the vertical tower flame spread facility. Some of the preliminary results in which the flow velocity is changed from +60 cm/s to -60 cm/s (wind opposed and wind-aided configurations) are reported here for the first time. The development of the object-oriented three dimensional computational code is also proceeding well. We have requested a no-cost extension of the grant. Hopefully, by the next reporting period we will have some new computational results to present.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Bhattacharjee S, Ayala R, Wakai K, Takahashi S. "Opposed-flow flame spread in microgravity-theoretical prediction of spread rate and flammability map." Proceedings of the Combustion Institute, 2005 Jan;30(2):2279-86. http://dx.doi.org/10.1016/j.proci.2004.08.020 , Jan-2005
Articles in Peer-reviewed Journals	Olson SL, Hegde U, Bhattacharjee S, Deering JL, Tang L, Altenrich RA. "Sounding rocket microgravity experiments elucidating diffusive and radiative transport effects on flame spread over thermally thick solids." Combustion Science and Technology, 2004 Apr;176(4):557-84. http://dx.doi.org/10.1080/00102200490276773 , Apr-2004
Presentation	Bhattacharjee, S., Paolini, C., Wakai, K., and and Takahashi, S. "Flammability Map for Microgravity Flame Spread" Strategic Research to Enable NASA's Exploration Missions Conference, NASA/TM-2004-213114, June, 2004 Jun-2004

Presentation	Christopher Paolini, Kyoung H. Yeo, and Subrata Bhattacharjee "An object oriented formulation for unsteady 3d heat transfer" Proceedings of CHT-04 ICHMT International Symposium on Advances in Computational Heat Transfer, April 2004 Apr-2004